

EXERCISE 1

a) $Y_i = \beta_1 + \beta_2 x_{i2} + \beta_3 x_{i3} + \beta_4 x_{i4} + \beta_5 x_{i5} + \varepsilon_i$

$i = 1, \dots, 13$ with $\varepsilon_i \sim N(0, \sigma^2)$ independent.

where x_{i2} = proportion of aluminum

x_{i3} = proportion of silicate

x_{i4} = proportion of aluminum-ferite

x_{i5} = proportion of silicate-bic

b) ① "std error" of $\hat{\beta}_2$ (aluminum)

$t^{obs} = 3.435$ (from the Table) is the observed value of the test

for testing $\begin{cases} H_0: \beta_2 = 0 \\ H_1: \beta_2 \neq 0 \end{cases}$

$$t^{obs} = \frac{\hat{\beta}_2 - 0}{\sqrt{\hat{Var}(\hat{\beta}_2)}} = \frac{0.9739}{0.2885} = 3.435 \Rightarrow \text{std error}(\hat{\beta}_2) = \frac{0.9739}{3.435} = 0.2885$$

② t statistic of aluminum-ferite

$$t^{obs} = \frac{\hat{\beta}_4 - 0}{\sqrt{\hat{Var}(\hat{\beta}_4)}} = \frac{-0.4974}{0.2751} = -1.808$$

the pvalue " $\Pr(>|t|)$ " is

$$\alpha^{obs} = \Pr_{H_0}(|T| > |t^{obs}|)$$

$$= 2 \Pr_{H_0}(T > |t^{obs}|) = 2 \Pr_{H_0}(T > |-1.808|)$$

where $T \stackrel{H_0}{\sim} t_8$ (Student's t with 8 = n-p d.o.f.)

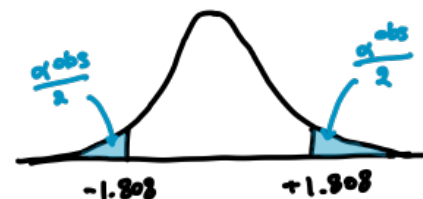
$$0.90 < \Pr(T \leq 1.808) < 0.95$$

$$-0.90 > -\Pr(T \leq 1.808) > -0.95$$

$$1 - 0.90 > 1 - \Pr(T \leq 1.808) > 1 - 0.95$$

$$0.10 > \Pr(T > 1.808) > 0.05$$

$$0.20 > 2 \Pr(T > 1.808) > 0.10 \Rightarrow \alpha^{obs} \in (0.10, 0.20)$$



③ estimate of the coeff. of silicate-bic

$$t^{obs} = \frac{\hat{\beta}_5}{\sqrt{\hat{Var}(\hat{\beta}_5)}}$$

$$-2.481 = t^{obs} = \frac{\hat{\beta}_5}{0.3214} \Rightarrow \hat{\beta}_5 = -2.481 \cdot 0.3214 = -0.7974$$

④ $R^2 = \frac{SSR}{SST} = \frac{\sum_{i=1}^{13} (\hat{y}_i - \bar{y})^2}{\sum_{i=1}^{13} (y_i - \bar{y})^2}$

we have the error sum of squares: $SSE = 49.378$

and the total sum of squares: $SST = 2715.76$

the deviance decomposition is $SST = SSE + SSR$

$$\Rightarrow SSR = 2715.76 - 49.378 = 2666.385$$

$$\text{Hence } R^2 = \frac{2666.385}{2715.76} = 0.9813$$

If I consider, for example, a fixed significance level $\alpha = 0.05$, the statistically significant variables are the ones for which $\alpha^{obs} < 0.05$, hence, the intercept, aluminum, and silicate-bic.

c) $\begin{cases} H_0: \beta_2 = \beta_3 = \beta_4 = \beta_5 = 0 \\ H_1: H_0 \text{ (at least one is } \neq 0) \end{cases}$

under H_0 the model is $Y_i = \beta_1 + \varepsilon_i \rightarrow \hat{\sigma}^2 = \frac{1}{13} \sum_{i=1}^{13} (y_i - \bar{y})^2 = \frac{1}{13} SST$

under H_1 the estimate of σ^2 is $\hat{\sigma}^2 = \frac{1}{13} \sum_{i=1}^{13} (y_i - \hat{y}_i)^2 = \frac{1}{13} SSE$

$$F = \frac{\frac{\hat{\sigma}^2 - \hat{\sigma}^2}{\hat{\sigma}^2} \cdot \frac{n-p}{p-1}}{\frac{SSE}{SST}} = \frac{SST - SSE}{SSE} \cdot \frac{13-5}{5-1} = \frac{SSR}{SSE} \cdot \frac{8}{4} \stackrel{H_0}{\sim} F_{4,8}$$

$$f^{obs} = \frac{2666.385}{49.378} \cdot 2 = 108.01$$

I reject H_0 if $f^{obs} > F_{4,8; 1-\alpha}$

here, I reject at all significance levels α

d) model B: $Y_i = \beta_1 + \beta_2 x_{i2} + \beta_5 x_{i5} + \varepsilon_i$

$$SSE_B = 74.762$$

$$\begin{cases} H_0: \beta_3 = \beta_4 = 0 \\ H_1: \text{at least one of } (\beta_3, \beta_4) \neq 0 \end{cases}$$

I use the statistic

$$F = \frac{\frac{\hat{\sigma}^2 - \hat{\sigma}^2}{\hat{\sigma}^2} \cdot \frac{n-p}{p-p_0}}{\frac{SSE_A}{SSE_B}} \stackrel{H_0}{\sim} F_{p-p_0, n-p}$$

$\hat{\sigma}^2$ is the estimate of σ^2 in model B

$$\hat{\sigma}^2 = \frac{1}{13} SSE_B$$

$\hat{\sigma}^2$ is the estimate of σ^2 in model A

$$\hat{\sigma}^2 = \frac{1}{13} SSE_A$$

$$p-p_0 = 5-3 = 2$$

$$n-p = 13-5 = 8$$

$$\Rightarrow F = \frac{SSE_B - SSE_A}{SSE_A} \cdot \frac{2}{2} \stackrel{H_0}{\sim} F_{2,8}$$

$$f^{obs} = \frac{74.762 - 49.378}{49.378} \cdot 4 = 2.056$$

I reject H_0 if $f^{obs} > F_{2,8; 1-\alpha}$

I do not reject H_0 for all usual α

I prefer model B since there is no evidence supporting the need to include x_3 and x_4 , and a parsimonious model is preferable.

e) $R_B^2 = \frac{SSR_B}{SST} = \frac{SST - SSE_B}{SST} = 1 - \frac{74.762}{2715.763} = 0.9725$

No, because the two models are nested and $R_A^2 \geq R_B^2$ by construction.

f) residuals vs fitted: e_i vs $\hat{y}_i$

we use it to check if the model's assumptions are satisfied, specifically orthogonality of the residuals and homoscedasticity.

A "good" model would produce a plot without particular patterns

Here, the plot does not highlight troubling behaviors.

normal q-q plot

used to check the normality assumption: empirical vs theoretical quantiles

If the hyp. is satisfied, points should lie on the line

Here, we have few data to derive accurate conclusions.

However, the data do not show troubling patterns.

EXERCISE 2

$(y_1, \dots, y_5)$ sample from $Y_i \sim \text{Pois}(e^{\beta_1})$ indep. $i = 1, \dots, 5$
 $(y_6, \dots, y_{10})$ sample from $Y_i \sim \text{Pois}(e^{\beta_1 + \beta_2})$ indep. $i = 6, \dots, 10$

a) • $Y_i \sim \text{Poisson}(\mu_i)$ $i = 1, \dots, 10$ indep.
 • $\eta_i = \beta_1 + \beta_2 x_i$ with $x_i = \begin{cases} 0 & i = 1, \dots, 5 \\ 1 & i = 6, \dots, 10 \end{cases}$
 • $\log \mu_i = \eta_i$

b) $f(y_i; \mu_i) = e^{-\mu_i} \mu_i^{y_i} \cdot \frac{1}{y_i!}$
 $f(y_1, \dots, y_{10}; \underline{\mu}) = \prod_{i=1}^{10} \left(e^{-\mu_i} \mu_i^{y_i} \frac{1}{y_i!} \right)$

$L(\underline{\mu}) \propto \prod_{i=1}^{10} e^{-\mu_i} \mu_i^{y_i}$ Likelihood

$\ell(\underline{\mu}) = \sum_{i=1}^{10} \{-\mu_i + y_i \log \mu_i\}$ loglikelihood. Since the parameters of interest are (β_1, β_2) and $\mu_i = e^{\beta_1 + \beta_2 x_i}$

$$\begin{aligned} \ell(\beta_1, \beta_2) &= \sum_{i=1}^{10} \left\{ -e^{\beta_1 + \beta_2 x_i} + y_i (\beta_1 + \beta_2 x_i) \right\} \leftarrow \text{loglikelihood of } (\beta_1, \beta_2) \\ &= -\sum_{i=1}^{10} e^{\beta_1 + \beta_2 x_i} + \beta_1 \sum_{i=1}^{10} y_i + \beta_2 \sum_{i=1}^{10} x_i y_i \\ &= -\sum_{i=1}^5 e^{\beta_1} - \sum_{i=6}^{10} e^{\beta_1 + \beta_2} + \beta_1 \sum_{i=1}^{10} y_i + \beta_2 \sum_{i=6}^{10} y_i \\ &= -5e^{\beta_1} - 5e^{\beta_1 + \beta_2} + \beta_1 \sum_{i=1}^{10} y_i + \beta_2 \sum_{i=6}^{10} y_i \end{aligned}$$

$$\ell_s(\beta_1, \beta_2) = \begin{cases} \frac{\partial \ell(\beta)}{\partial \beta_1} = -5e^{\beta_1} - 5e^{\beta_1 + \beta_2} + \sum_{i=1}^{10} y_i \\ \frac{\partial \ell(\beta)}{\partial \beta_2} = -5e^{\beta_1 + \beta_2} + \sum_{i=6}^{10} y_i \end{cases} \quad \text{score function}$$

$$\ell_{st}(\beta_1, \beta_2) = \begin{bmatrix} -5e^{\beta_1} - 5e^{\beta_1 + \beta_2} & -5e^{\beta_1 + \beta_2} \\ -5e^{\beta_1 + \beta_2} & -5e^{\beta_1 + \beta_2} \end{bmatrix} = -5 \begin{bmatrix} e^{\beta_1} + e^{\beta_1 + \beta_2} & e^{\beta_1 + \beta_2} \\ e^{\beta_1 + \beta_2} & e^{\beta_1 + \beta_2} \end{bmatrix}$$

observed info is $j(\underline{\beta}) = -\ell_{st}(\underline{\beta})$

The MLE can be found by noticing that

for sample 1 $(y_1, \dots, y_5) = \underline{y}_1$ the expected value is $E[Y_i] = \mu_1 = e^{\beta_1}$
 for sample 2 $(y_6, \dots, y_{10}) = \underline{y}_2$ the expected value is $E[Y_i] = \mu_2 = e^{\beta_1 + \beta_2}$

The function from (μ_1, μ_2) to (β_1, β_2) is bijective: $(\beta_1, \beta_2) = f(\mu_1, \mu_2)$

I can obtain the MLE $(\hat{\mu}_1, \hat{\mu}_2)$ and obtain $(\hat{\beta}_1, \hat{\beta}_2) = f(\hat{\mu}_1, \hat{\mu}_2)$

the MLE of μ_1 and μ_2 are

$$\hat{\mu}_1 = \bar{y}_1 = \frac{1}{5} \sum_{i=1}^5 y_i \quad \text{sample mean of } \underline{y}_1$$

$$\hat{\mu}_2 = \bar{y}_2 = \frac{1}{5} \sum_{i=6}^{10} y_i \quad \text{sample mean of } \underline{y}_2$$

$$\Rightarrow \begin{cases} \hat{\mu}_1 = e^{\hat{\beta}_1} \\ \hat{\mu}_2 = e^{\hat{\beta}_1 + \hat{\beta}_2} \end{cases} \Rightarrow \begin{cases} \hat{\beta}_1 = \log \hat{\mu}_1 = \log \bar{y}_1 \\ \hat{\beta}_1 + \hat{\beta}_2 = \log \hat{\mu}_2 \Rightarrow \hat{\beta}_2 = \log \hat{\mu}_2 - \hat{\beta}_1 = \\ = \log \hat{\mu}_2 - \log \hat{\mu}_1 \end{cases}$$

$$\text{hence } \begin{cases} \hat{\beta}_1 = \log \bar{y}_1 \\ \hat{\beta}_2 = \log \bar{y}_2 - \log \bar{y}_1 = \log \frac{\bar{y}_2}{\bar{y}_1} \end{cases} \quad \left(\begin{array}{l} e^{\hat{\beta}_1} = \bar{y}_1 \\ e^{\hat{\beta}_2} = \frac{\bar{y}_2}{\bar{y}_1} \end{array} \right)$$

Alternatively, one can solve the likelihood equations

$$\begin{cases} -5e^{\beta_1} - 5e^{\beta_1 + \beta_2} + \sum_{i=1}^{10} y_i = 0 & \rightarrow 5e^{\beta_1} = \sum_{i=1}^5 y_i - \sum_{i=6}^{10} y_i \rightarrow e^{\beta_1} = \frac{1}{5} \left(\sum_{i=1}^5 y_i - \sum_{i=6}^{10} y_i \right) = \bar{y}_1 \\ -5e^{\beta_1 + \beta_2} + \sum_{i=6}^{10} y_i = 0 & \rightarrow e^{\beta_1 + \beta_2} = \frac{\sum_{i=6}^{10} y_i}{5} = \bar{y}_2 \end{cases}$$

$$c) \begin{bmatrix} \hat{\beta}_1 \\ \hat{\beta}_2 \end{bmatrix} \sim N_2 \left(\begin{bmatrix} \beta_1 \\ \beta_2 \end{bmatrix}, j(\hat{\underline{\beta}})^{-1} \right)$$

the marginal $\hat{\beta}_1 \sim N(\beta_1, [j(\hat{\underline{\beta}})^{-1}]_{(1,1)})$
 ↪ element in position (1,1)

d) we already noticed that

for sample 1 $E[Y_i] = \mu_1 = e^{\beta_1}$

for sample 2 $E[Y_i] = \mu_2 = e^{\beta_1 + \beta_2}$

$$\frac{\mu_2}{\mu_1} = \frac{e^{\beta_1 + \beta_2}}{e^{\beta_1}} = e^{\beta_2} \Rightarrow \beta_2 = \log \frac{\mu_2}{\mu_1} = \log \frac{E[Y_i | \text{sample 2}]}{E[Y_i | \text{sample 1}]}$$

Hence β_2 represents the log of the ratio between the means of the two samples.

Or, noting that $\mu_2 = e^{\beta_2} \cdot \mu_1$

the mean of sample 2 is obtained by multiplying the mean of sample 1 of a coefficient e^{β_2}

e) The saturated model is a model with n parameters $\mu_1, \dots, \mu_n$.

In this case, I have one parameter for each observation and the estimates are

$$\hat{\mu}_i = y_i \quad \forall i$$

Hence the max of the loglikelihood is

$$\ell^0(\text{saturated}) = \sum_{i=1}^{10} \{-\hat{\mu}_i + y_i \log \hat{\mu}_i\} = \sum_{i=1}^{10} \{-y_i + y_i \log y_i\}$$